

Cryptography and Network Security

Technical Report

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1 Introduction

This project demonstrates practical applications of cryptography, data integrity verification, risk assessment and network access control in an authorised laboratory environment.

The main security objectives of the project are to protect sensitive information, detect unauthorised modification of data and control access to network services.

The project includes an encryption and decryption programme, an integrity verification function using a cryptographic hash, a risk assessment and a firewall access-control configuration.

2 Risk Assessment

2.1 Assets, Vulnerabilities and Consequences

The following assets and vulnerabilities were identified during the risk assessment.

Asset	Vulnerability	Possible Consequence
Student records	Weak staff passwords	An attacker could gain unauthorised access to student information.
Central records server	Outdated software	An attacker could exploit known weaknesses and compromise the server.
Files transferred between campuses	Unencrypted file transfers	Sensitive information could be intercepted during transmission.

2.2 Risk Ranking

Three major risks were identified and ranked according to likelihood and impact.

Risk	Likelihood	Impact	Reason
Unauthorised access to student records	High	High	Weak staff passwords increase the possibility of account compromise, and access to student records could expose sensitive information.

Exploitation of outdated software	out-Medium	High	Outdated software may contain exploitable weaknesses that could allow an attacker to compromise the central server.
Interception of transferred files	trans-Medium	High	Unencrypted transfers can expose information while files are being transferred between campuses.

2.3 Recommended Controls

The following controls were recommended to reduce the identified risks.

Risk	Recommended Control
Unauthorised access to student records	Enforce strong staff passwords and multi-factor authentication.
Exploitation of outdated software	Regularly update and patch the server software.
Interception of transferred files	Use encrypted file transfers between the campuses.

3 Encryption and Decryption

Encryption is used to protect information by converting readable plaintext into ciphertext. The ciphertext should not reveal the original information to an unauthorised person.

The general process used in the project is:

Plaintext → Encryption → Ciphertext

Decryption reverses this process and converts the ciphertext back into the original plaintext.

Ciphertext → Decryption → Original Plaintext

The programme was tested to verify that information encrypted by the programme could subsequently be decrypted correctly. A successful decryption should reproduce the original plaintext.

3.1 Encryption Test Evidence

The encryption test follows these steps:

1. Input plaintext into the programme.
2. Encrypt the plaintext.
3. Observe the resulting ciphertext.
4. Decrypt the ciphertext.
5. Compare the decrypted text with the original plaintext.

The expected result is that the decrypted text matches the original plaintext.

4 Integrity Verification

Integrity verification determines whether information has been changed after it was originally created or transmitted.

The project uses a cryptographic hash function for integrity checking. A hash converts data into a fixed-length hash value.

The process can be represented as:

Original Data → Hash Function → Hash Value

If the original data is modified, calculating the hash again should produce a different hash value.

4.1 Integrity Test

The integrity test consists of the following procedure:

1. Calculate the hash of the original data.
2. Store or record the resulting hash value.
3. Modify the data.
4. Calculate the hash again.
5. Compare the two hash values.

If the hash values differ, the data has been modified.

The test therefore demonstrates that cryptographic hashing can be used to detect changes to information.

5 Firewall Configuration

A firewall was designed for an authorised laboratory environment to demonstrate network access control.

The laboratory network configuration is shown below.

Component	Configuration
Student Records Server	192.168.10.10
Guest Network	192.168.20.0/24
Authorised Staff Network	192.168.30.0/24
Protected Service	SSH
Protocol	TCP
Port	22

Table 4: Laboratory firewall environment

The firewall policy consists of three main access-control requirements.

5.1 Guest Network Blocking

Access from the guest network to the Student Records Server is blocked. This prevents guest devices from accessing the protected server.

5.2 Authorised Staff Access

The authorised staff network is permitted to access the protected SSH service on TCP port 22.

5.3 Other Inbound Access

Inbound access to the protected SSH service from sources outside the authorised staff network is blocked.

The general policy is therefore:

Source	Destination	Action
Guest Network	Student Records Server	BLOCK
Staff Network	SSH Port 22	ALLOW
Other Sources	SSH Port 22	BLOCK

The firewall configuration is documented in the project file `firewall.conf`.

6 Firewall Test Procedure and Results

Three test cases were defined to verify the firewall access-control policy. One test checks permitted access and two tests check blocked access.

6.1 Test 1 – Authorised Staff Connection

The first test verifies that an authorised staff computer can connect to the SSH service.

```
nc -vz 192.168.10.10 22
```

Expected outcome: The connection should be permitted because the source belongs to the authorised staff network.

Simulated laboratory result:

```
Connection to 192.168.10.10 port 22 [tcp/ssh] succeeded!
```

The result indicates that the authorised connection was permitted.

6.2 Test 2 – Guest Network Connection

The second test verifies that a guest network connection is blocked.

```
nc -vz 192.168.10.10 22
```

Expected outcome: The connection should be blocked because the source belongs to the guest network.

Simulated laboratory result:

```
nc: connectx to 192.168.10.10 port 22 failed: Operation timed out
```

The result indicates that the guest connection was blocked.

6.3 Test 3 – Unauthorised Inbound Connection

The third test verifies that another unauthorised source cannot access the protected SSH service.

```
nc -vz 192.168.10.10 22
```

Expected outcome: The connection should be blocked because the source is outside the authorised staff network.

Simulated laboratory result:

```
nc: connectx to 192.168.10.10 port 22 failed: Operation timed out
```

The result indicates that the unauthorised connection was blocked.

Test	Source	Expected	Result
1	Authorised Staff	ALLOW	PASS
2	Guest Network	BLOCK	PASS
3	Other Network	BLOCK	PASS

Table 5: Firewall test summary

6.4 Firewall Test Summary

Note: The firewall results presented in this report are simulated laboratory results because no specific network addresses or service were supplied.

7 Conclusion

The project demonstrated several important network security principles.

The risk assessment identified weaknesses involving weak passwords, outdated software and unencrypted file transfers. Appropriate controls were recommended, including strong passwords, multi-factor authentication, regular software updates and encrypted file transfers.

The cryptography component demonstrated encryption and decryption for protecting information. The integrity component demonstrated the use of cryptographic hashing to detect changes to data.

The firewall component demonstrated network access control by allowing authorised staff access to a protected service while blocking guest and other unauthorised connections.

Together, these components demonstrate how confidentiality, integrity, risk management and network access control can be applied as part of a basic information security solution.

8 GitHub Repository

The complete project, including the source code, risk assessment, firewall configuration, test evidence, README documentation and LaTeX report, is maintained in the GitHub repository.

Repository URL:

<https://github.com/fabuwase/cryptography-network-security-exam>